Inborn Errors of Metabolism 2

Gada jadkarim

Clinical and Biochemical Geneticist

King Abdulaziz University

Objectives

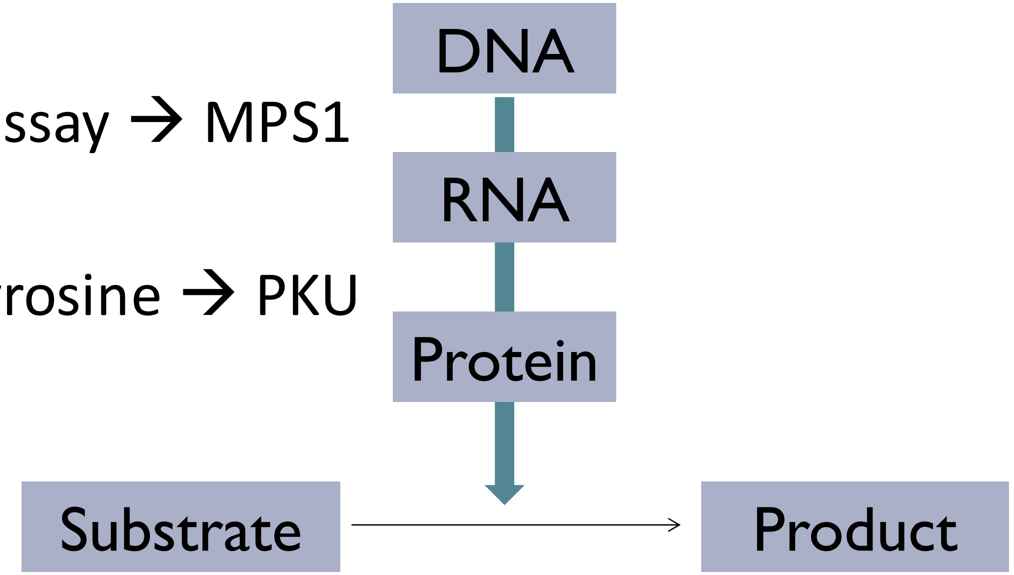
- Identify types of samples used for diagnosis of IEMs
- Enumerate the 1st and 2nd line tests used in IEM
- Interpret the results of the tests and mention their significance
- Identify follow-up steps to the investigations ordered

Indications of Laboratory

- To aid in the diagnosis of IEMs
- Monitoring of disease control
- Screening for disease complications

Levels of investigations and examples

- DNA
  - Molecular tests → *OTC* gene sequencing
- RNA
  - Mainly done for research purposes, not well incorporated into clinical practice
- Protein
  - Enzyme Assays → α-L-iduronidase enzyme assay → MPS1
- Substrate/Product
  - Plasma amino acids → Phenylalanine and Tyrosine → PKU
- Organ damage
  - Liver function tests → GSD Type 1a



Samples used in the diagnosis of IEMs

- Dried blood spots
- Peripheral Blood
- Urine
- Cerebrospinal Fluid
- Other tissues → Skin, hair, muscle and liver

First line vs second line investigations

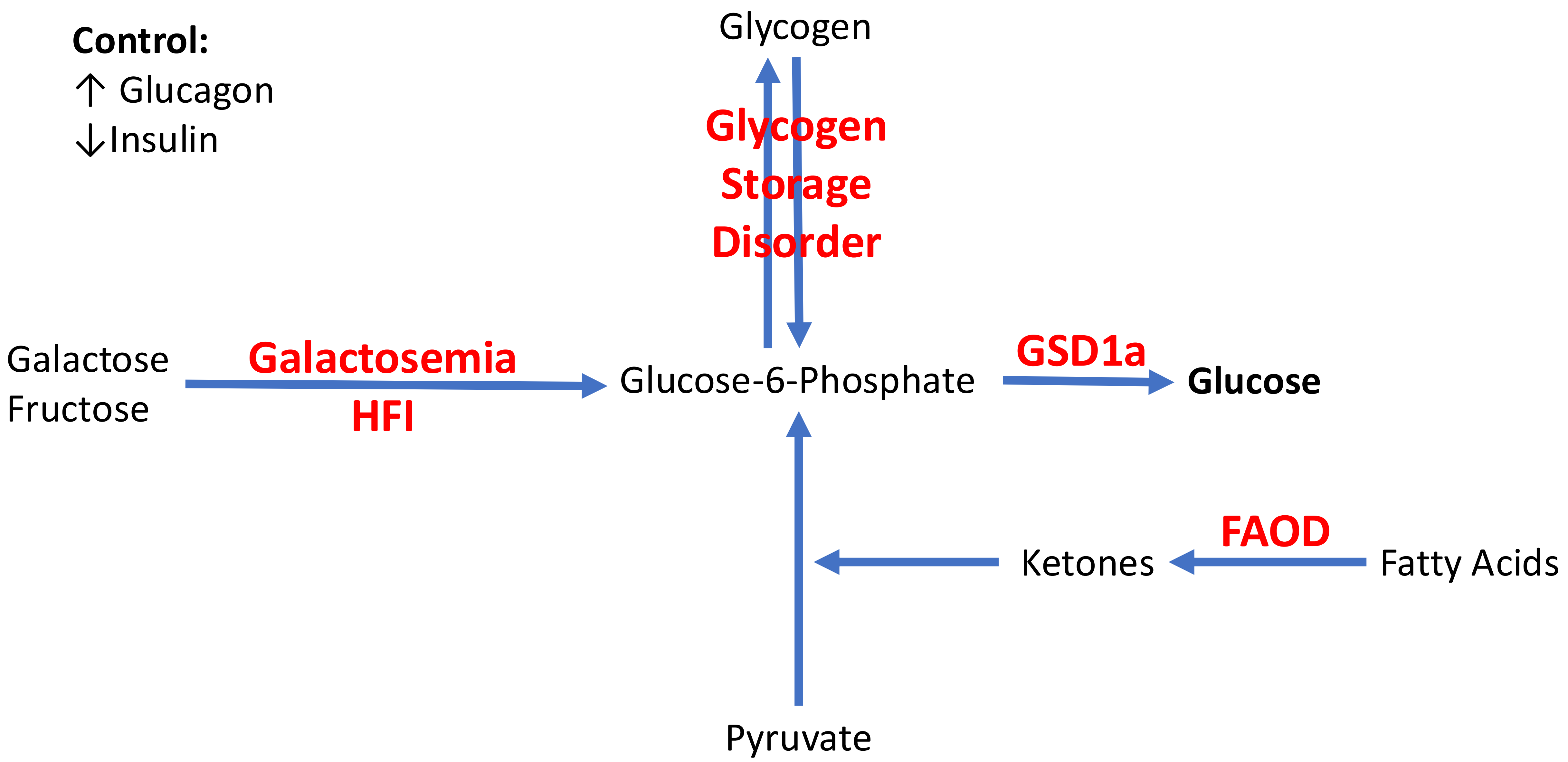
First line	Second line
Non-specific	Specific
Screening	Can be diagnostic
Fast turn around time for results	Takes days to weeks to obtain results
Cheap	Expensive
Readily Available	Requires Special Coordination
Easy to interpret	Can be difficult to interpret

First Line Investigations

First Line Investigations

- Complete blood count, blood film
- Blood gas (acidosis or alkalosis)
- Ammonia level
- Lactic acid
- Glucose
- Liver enzymes and function
- Muscle enzymes (CK)
- Urine ketone
- Urine Reducing substance

Glucose


Control:
↑ Glucagon
↓Insulin
Glycogen
Glycogen Storage Disorder
Galactose
Fructose
Galactosemia
HFI
Glucose-6-Phosphate
GSD1a
Glucose
FAOD
Ketones
Fatty Acids
Pyruvate

Hypoglycemia

- ↓ blood Glucose
  - <2.2 mmol/L (40 mg/dl) on first day of life
  - 2.2–2.8 mmol/L (40-50 mg/dl) after 24 hours of age

Hypoglycemia

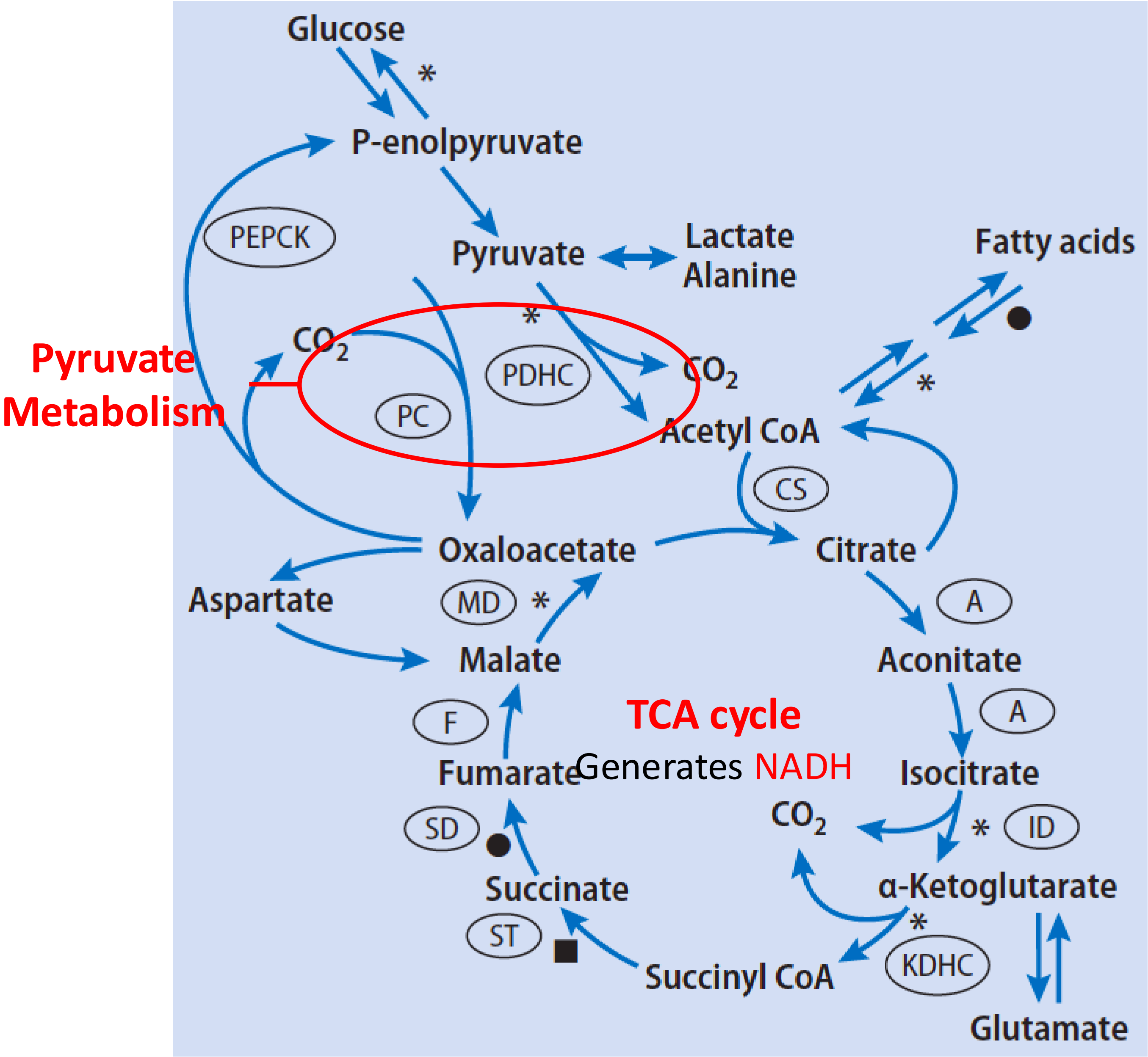
Ketotic

- Glycogen storage diseases
- Galactosemia
- Hereditary Fructose intolerance (HFI)

Non/hypo-ketotic

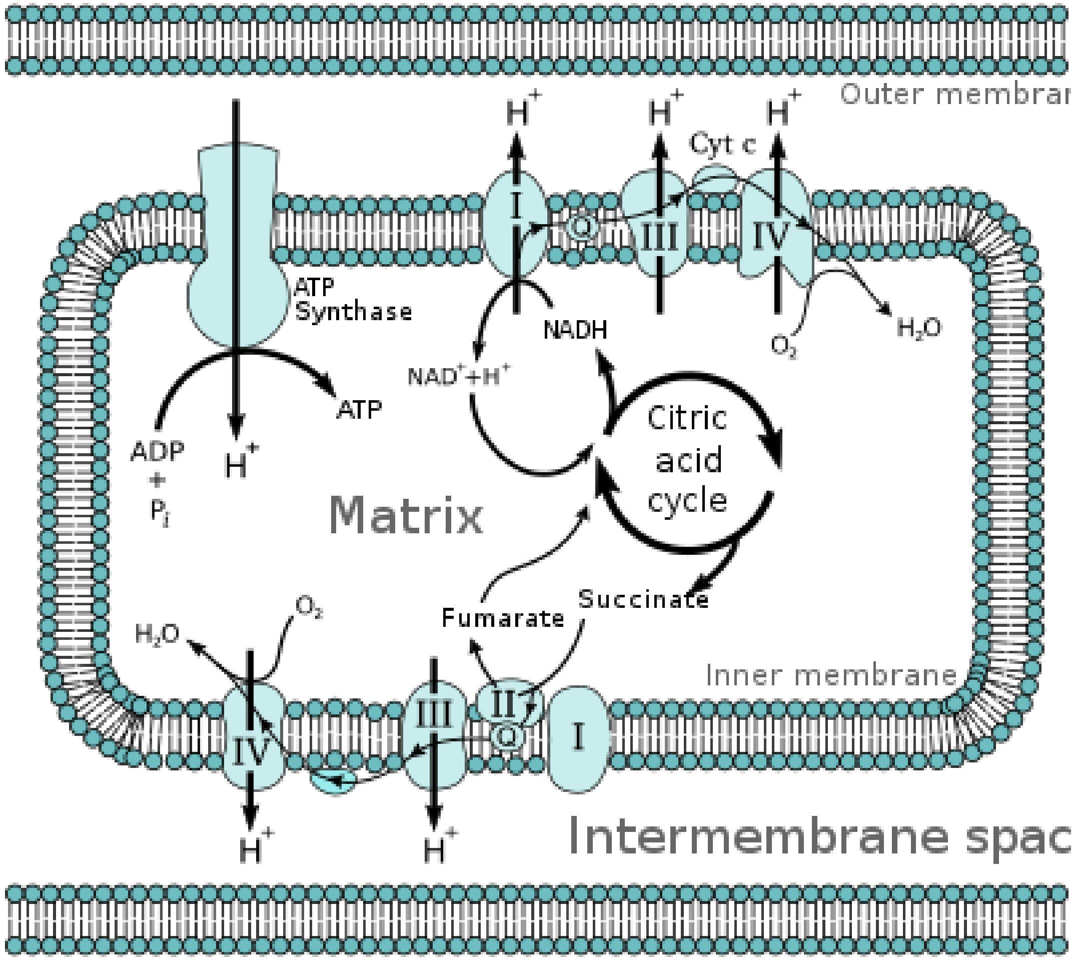
- Fatty acid oxidation defect (FAOD)
- Hyperinsulinism

Lactic Acid



*De Meirleir LJ. et al. (2016). Disorders of Pyruvate Metabolism and the Tricarboxylic Acid Cycle. In Saudubray J-M. et al, *Inborn Metabolic Diseases*. p.185-199.

NADH utilized by RCE to generate ATP



'Electron transport chain' (2021) Wikipedia. Available at: https://en.wikipedia.org/wiki/Electron_transport_chain (Accessed: 5 January 2021).

Lactic Acidosis

- Accumulation of Blood Lactate
  - Leads to metabolic acidosis → High anion gap
- Normal lactate <2.0 mmol/L

IEM causes

- Primary
  - Pyruvate metabolic defects
  - TCA cycle defects
  - Oxidative phosphorylation defects
- Secondary
  - Gluconeogenesis defects
  - GSD1a
  - Organic acidemias
  - Fatty acid oxidation disorders

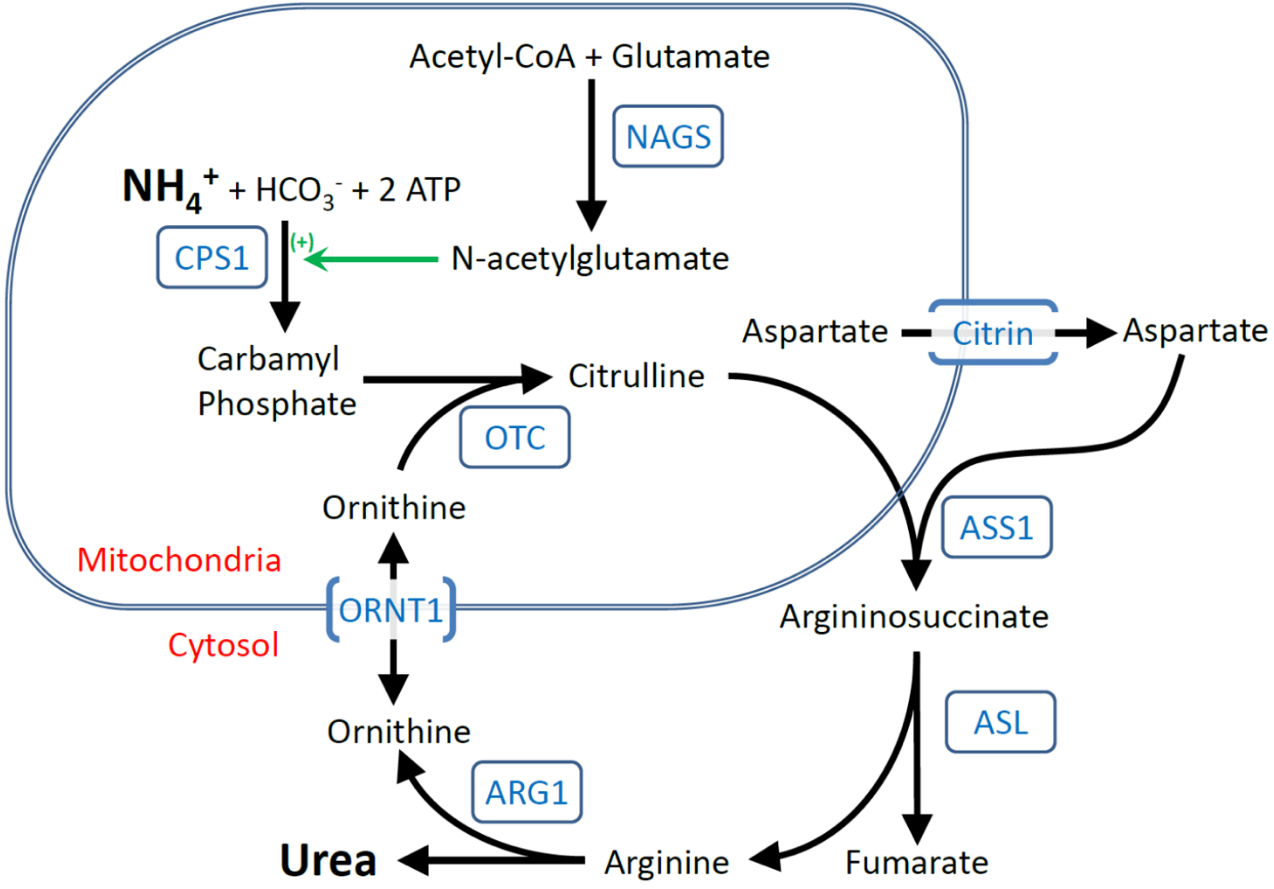
Non-IEM causes → More common

- Sample Hemolysis → Most common causes
- Circulatory collapse
- Hypoxic insult
- Diarrhea
- Sepsis
- Hypoventilation
- Hepatic failure

Ammonia

Nitrogen is the waste product of all amino acids → In the form of Ammonia

- Muscle turnover/breakdown
- Protein



*Sanchez Russo RL & Wilcox WR.(2021) Amino Acid Metabolism. In Pyeritz RE et al. Emery and Rimoin's Principles and Practice of Medical Genetics and Genomics: Metabolic Disorders. Elsevier. 49-104

Hyperammonemia

- Elevated blood ammonia (Normal <50mmol/L)
- Causes:
  - IEM
    - Primary
      - Urea Cycle Defect
    - Secondary
      - FAOD
      - Organic Acidemia
      - Oxidative phosphorylation defects
- Non-IEM
  - Drug-related (valproate)
  - Liver failure (Acute or Chronic)
  - Massive tissue necrosis
  - Overgrowth of bowel flora
  - Transient hyperammonemia in the newborn

Second Line Investigations

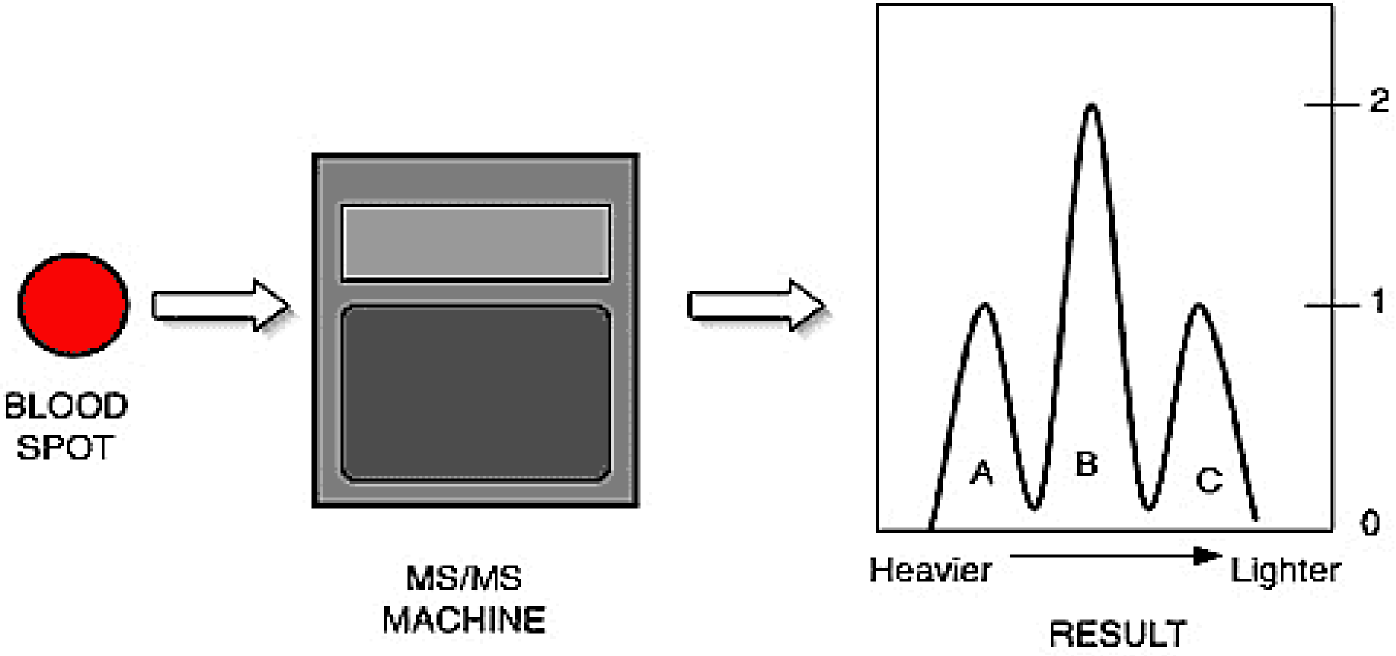
Second Line Investigations

- Plasma, Urine & CSF amino acid
- Acylcarnitine (Tandem MS)
- Plasma Homocysteine
- Plasma Very long chain fatty acid
- Urine organic acid
- Urine mucopolysaccharides
- Urine Oligosaccharides
- Specific enzyme assay (Blood, tissue)

Tandem Mass Spectrometry (MS/MS)

- A method of measuring analytes by both mass and structure.
- Steps:
    - Ionization of the compounds
    - Selects ions according weight
    - Fragmentation
    - Detection
    - Generates results

MS/MS



What does MS/MS detect?

- Amino acids (Plasma, Urine, CSF)
- Acylcarnitine (Free and bound)
- Organic acid
- Fatty acid metabolites

TABLE 1. Metabolic disorders detectable in newborns aged 1–5 days by using tandem mass spectrometry

Disorder	Primary metabolic indicator
Amino Acids	
Phenylketonuria	Phe
Maple syrup urine disease	Leu/Ile, Val
Homocystinuria (cystathione synthase deficiency)	Met
Hypermethioninemia	Met
Citrullinemia	Cit
Argininosuccinic aciduria	Cit
Tyrosinemia, type I	Tyr
Fatty Acids	
Medium-chain acyl-CoA dehydrogenase deficiency	C8, C10, C10:1, C6
Very-long–chain acyl-CoA dehydrogenase deficiency	C14:1, C14, C16
Short-chain acyl-CoA dehydrogenase deficiency	C4
Multiple acyl-CoA dehydrogenase deficiency	C4, C5, C8:1, C8, C12, C14, C16, C5DC
Carnitine palmitoyl transferase deficiency	C16, C18:1, C18
Carnitine/acylcarnitine translocase defect	C16, C18:1, C18
Long-chain hydroxy acyl-CoA dehydrogenase deficiency	C16OH, C18:1OH, C18OH
Trifunctional protein deficiency	C16OH, C18:1OH, C18OH
Organic Acids	
Glutaric acidemia, type I	C5DC
Propionic acidemia	C3
Methylmalonic acidemia	C3
Isovaleric acidemia	C5
3-hydroxy-3-methylglutaryl CoA lyase deficiency	C5OH
3-methylcrotonyl CoA carboxylase deficiency	C5OH

Plasma Amino Acids

Amino acid	Value	Disorder
Arginine	High	Arginase def. (UCD)
Citrulline	High	Citrullinemia (UCD)
Glycine	High	Hyperglycinemia
Leucine	High	Maple syrup urine disease
Isoleucine	High	Maple syrup urine disease
Valine	High	Maple syrup urine disease
Phenylalanine	High	Phenylketonuria
Tyrosine	High	Tyrosinemia
Tyrosine	Low	Phenylketonuria
Homocysteine	High	Homocystinuria

Case

- 7 days old female neonate presents with:
  - Vomiting, lethargy, and seizure on breast feeding
  - Family history : positive consanguinity
- On examination:
  - Ill looking, tachypneic, with hypotonia.

- What labs would you order?

1st line investigations

- Complete blood count, blood film
- Blood gas
- Ammonia level
- Lactic acid
- Glucose
- Liver enzymes and function
- Muscle enzymes (CK)
- Urine ketone
- Urine Reducing substance

1^{st} line investigations

- Complete blood count, blood film
  - CBC normal → Blood film not required
- Blood gas (Normal PH 7.35-7.45)
  - Respiratory alkalosis
- Ammonia level (Normal <100 µmol/L)
  - 1267 µmol/L
- Lactic acid (Normal <2)
  - Normal
- Glucose
  - 4.0 mmol/L
- Liver enzymes and function
  - Mildly elevated
- Muscle enzymes (CK)
  - Mildly elevated
- Urine ketone
  - Absent
- Urine Reducing substance
  - Absent

Hyperammonemia - Differential Diagnosis

- IEM
  - Primary
    - Urea cycle
  - Secondary
    - Organic acidemias
    - Fatty acid oxidation defects
    - Respiratory chain defects
- Non-IEM
  - Drug-related
  - Liver Failure
  - Massive tissue necrosis
  - Overgrowth of bowel flora
  - Liver Cirrhosis
  - Transient hyperammonemia in the newborn

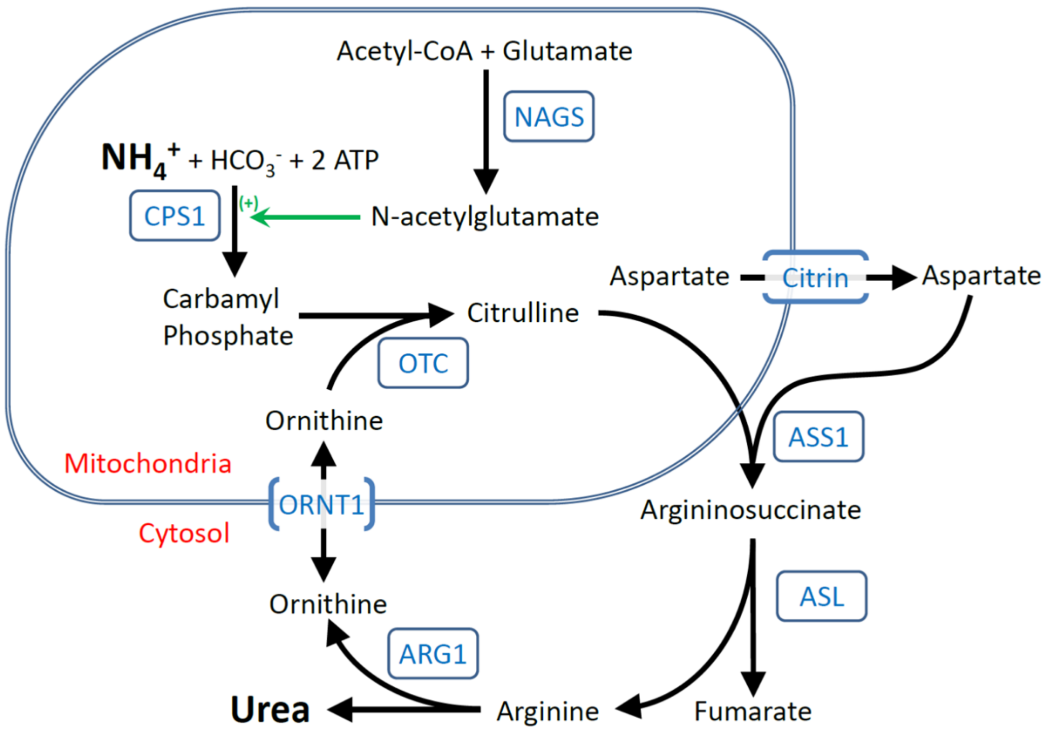
Hyperammonemia - Differential Diagnosis

- IEM
  - Primary
    - Urea cycle
  - Secondary
    - Organic acidemias
    - Fatty acid oxidation defects
    - Respiratory chain defects

- What tests to order?

2nd Tier tests

- Plasma amino acids
  - Elevated Glutamine
  - Elevated Citrulline
  - Reduced Arginine
- Urine Orotic acid
  - Elevated
- Arginosuccinate synthase
  - Citrullinemia type 1



*Sanchez Russo RL & Wilcox WR.(2021) Amino Acid Metabolism. In Pyeritz RE et al. Emery and Rimoin's Principles and Practice of Medical Genetics and Genomics: Metabolic Disorders. Elsavier. 49-104